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Impact of data visualization on decision-making and its implications for public health practice: a systematic literature review

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ABSTRACT

Data visualization tools have the potential to support decision-making for public health professionals. This review summarizes the science and evidence regarding data visualization and its impact on decision-making behavior as informed by cognitive processes such as understanding, attitude, or perception.

An electronic literature search was conducted using six databases, including reference list reviews. Search terms were pre-defined based on research questions.

Sixteen studies were included in the final analysis. Data visualization interventions in this review were found to impact attitude, perception, and decision-making compared to controls. These relationships between the interventions and outcomes appear to be explained by mediating factors such as perceived trustworthiness and quality, domain-specific knowledge, basic beliefs shared by social groups, and political beliefs.

Visualization appears to bring advantages by increasing the amount of information delivered and decreasing the cognitive and intellectual burden to interpret information for decision-making. However, understanding data visualization interventions specific to public health leaders' decision-making is lacking, and there is little guidance for understanding a participant's characteristics and tasks. The evidence from this review suggests positive effects of data visualization can be identified, depending on the control of confounding factors on attitude, perception, and decision-making.

KEYWORDS

Data visualization; decision-making; public health practice; local health department; public health informatics

Introduction

Data visualization refers to visual representations such as graphs and maps used to understand, communicate, and support collaborative engagement and decision-making with data and information.[1–4] Data visualizations can help people understand data, gain insight, answer specific questions, and discover underlying facts.[1,3–5] Data visualization tools have the potential to support public health practice by answering questions that previously were unanswerable, such as, what makes certain areas in which a population lives, or certain demographic groups within a population healthier than others.

Public health leaders in state and local health departments are expected to make the best possible decisions for resource allocation and public health planning. They are to do this based on their understanding of what factors influence which population and what intervention works to address health issues

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in specific populations and based on available scientific evidence through systematic uses of data and information systems.[6(p177),7,8] Making the best decisions regarding prevention programs and policies is crucial when considering their broad impact on various populations.[9] Since even before the 2008 U.S. economic recession, local health departments (LHDs) have faced severe cuts to budgets, programs, and staff,[10] and have desired a means to be more strategic in allocating limited funding to effective programs.[9]

In recent years, numerous studies from various fields, especially computer science and information science, have developed data visualization tools and described the utility and usability of these tools in their field. However, few studies have examined the impacts of data visualization tools in public health practice. Also, only a few studies have used formal experimental designs (e.g., randomized controlled trials) to examine the impact of data visualization on human understanding, perception, or behavior related to decision-making.

This systematic literature review assessed the current science and evidence regarding data visualization and its impact on decision-making-related behavior as informed by cognitive processes such as understanding, attitude, or perception; and identified key elements of appropriate study designs for testing this impact. The objective of this review was to identify theories, methods, and interventions used for testing data visualization, as well as primary and secondary outcomes of the effect of data visualization and mediating and moderating factors known to interfere with the impact of data visualization. While the articles included in this review were not entirely from the field of public health, our focus was to draw out related implications for public health professionals in this review.

Methods

We conducted this systematic literature review using guidance from the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) and the Center for Reviews and Dissemination (CRD) 's book "Systematic Reviews."[11,12]

Eligibility criteria

Types of patients/populations/people being exposed to the data visualizations

To fully capture studies related to our focus – the impacts of data visualization – we expanded our criteria to include studies examining the impacts of data visualization on any population, even though our initial target population was public health professionals from state and local health departments. Articles were excluded if they were related to animals or plants.

Types of studies/interventions

We included only experimental and quasi-experimental studies that were the most appropriate for identifying causal effects, given our focus on impact.[13] Articles were included if 1) they described a behavioral, social, or organizational intervention using data visualization, 2) their main aims were assessing the use of data visualization, 3) they described a method of measuring the impacts of data visualization, and 4) they clearly described the study's objectives, methods, and findings. Articles were excluded if 1) they were books, book chapters, magazine review articles, systematic reviews, or commentaries, 2) the full text of the articles could not be retrieved, or 3) they were unpublished research reports. There were many articles on data visualization from biochemistry, genomics, immunology, oncology, pathology, and nutrition specific to microscopic biomedical research. We also did not include these articles because our study focused on how visual displays that use individual, population, or organizational-level data affect human decision-making.

Data visualization was defined for this review as any visual display (e.g., bars, pie charts, line graphs) used for presenting data and information.

Types of outcomes

We did not include articles that only described the process of developing a visualization tool or that used visualization to describe a methodology used for statistical modeling because the focus of our review was experimental studies that measured the impact of visualization on cognitive and behavioral changes.

Search strategy and information sources

With the help of librarians that specialize in health science and data science, we conducted an electronic literature search in February 2021 using PubMed, CINAHL Plus with full text, Web of Science, IEEE Xplore, PAIS international, and Academic Search Complete, which are databases frequently used in the health sciences.[14] Then, we expanded the search to other disciplinary databases relevant to computer and information science where data sciences are actively studied.[15] To make the search more comprehensive, we also reviewed reference lists from searched articles that could be related to our research questions.[16]

The search strategy was developed by defining two main concepts from our research questions.[11] The two search concepts included: 1) "data visualization" as an intervention and 2) "decision-making related cognitive and behavior change" as an outcome. In addition to the two concepts, a third concept, "community and public health systems," as an intervention setting was used specifically for the PubMed and Web of Science search. For searches other than PubMed and Web of Science, including all three concepts in a database search did not generate any results because the search terms were too restrictive.

We limited our search to articles published between 2008 and 2020, peer-reviewed, written in English, and involving studies with humans. We did not have any restrictions such as education, age, or geographic location to maximize the chance of finding relevant articles focusing on the impacts of data visualization.

Selection of studies and data extraction

Searched articles were screened by an author (S.P) for their titles and abstracts for inclusion in the review based on the eligibility criteria. The full-text review for the final selection was independently conducted by two of the authors (S.P and M.S) using an Excel spreadsheet that listed inclusion and exclusion criteria. When there was any uncertainty or discrepancy about the inclusion and exclusion of the screened studies, the two authors discussed discrepancies until consensus was reached. Data from the included articles were extracted by a researcher (S.P) into a data extraction spreadsheet developed to collect all the relevant study data. Data extracted included information on the type of study design, aims, sample sizes, study populations, theories or framework used in the studies, study design and setting, intervention characteristics (e.g., number of participants, dosage, control group), outcome measures related to cognitive and behavioral changes, and mediating and moderating variables.

Assessment of quality and risk of bias

Retrieved articles were assessed for their quality and risk of bias. We assessed the articles according to the following criteria adapted from the Critical Appraisal Skills Programmes checklist[17]: 1) if the study designs were quasi-experimental or experiments used for evaluating the impact of data visualization tools as interventions related to cognitive and behavioral changes, 2) if the methods used to measure the impact of visualization were quantitative, 3) if the confounding factors were measured and controlled for their possible effect in the analysis, and 4) if the sampling and randomization methods were clearly stated.

Any uncertainty about the level of bias attributed to an individual study was discussed within the study team until a consensus was reached regarding inclusion.

Data synthesis

We used narrative synthesis with tabular presentations for evidence synthesis. Narrative synthesis can be used for qualitatively synthesizing findings from multiple studies when the quantitative outcomes of selected studies were not in comparable formats.[18,19(p135)]

Results

A total of 1140 articles were retrieved from electronic database searches, in addition to three additional articles identified by reviewing reference lists of articles. Combining the results of all searches and removing duplicates (n = 20) yielded 1123 articles. After reviewing the title and abstract of the articles based on eligibility criteria, we retained 80 articles for full-text review. Of these, we excluded 64 additional articles because 1) they focused only on the process of developing data visualization tools (e.g., appropriateness of color and size of elements of visualization), 2) they were a review or commentary article, or 3) they did not use data visualization as an intervention. A flow diagram (Figure 1) shows the screening process and reasons for exclusion based on eligibility assessment. We included the remaining 16 articles in the final qualitative synthesis. Ten studies' settings were located in the U.S., two studies were from Taiwan,[20,21] and others were from the Netherlands,[22] Spain,[23] Sweden,[24] and Switzerland.[25] We organized results by 1) theories used to explain how data visualization influences behavior, 2) study designs and intervention used for testing data visualization, and 3) outcome measures, including mediating and moderating factors used for their analysis of the impact of data visualization. Each of these elements is depicted in Figure 2.

Theory used for data visualization experiments

Most of the papers did not explicitly describe theories specifically related to data visualization in their experiments but described specific theories related to their topic rather than about data visualization. For example, a study by Pandey and colleagues adopted a theoretical model of persuasion, the Elaboration Likelihood Model, which described the dual process (central route and peripheral route) of a person processing stimuli that would influence attitude change. The process partially depended on the persons' ability to process the message without cognitive burden.[26,27] The authors used this theory to structure their experiment to measure the persuasive power of data visualization on controversial social issues. Regarding hypotheses, five studies started from the assumption that visual displays help increase understanding of the problem by integrating large amounts of information from multiple sources while reducing cognitive burden, resulting in effective and efficient information-based decision-making regarding complex real-world problems.[28–31]

Study designs, participants and interventions used for testing data visualization

Design

Descriptive information about designs and interventions of included studies is reported in Table 1. Comparing two groups assigned as the treatment group and the control group for treatment effects was the most common design (n = 14, 87.5%). Three studies compared the pre-and post-effects of treatment.[22,27,28] Among experimental studies, nine studies reported that they used randomization for assigning their participants, but only two studies reported the method of randomization.[21,31] Four other studies did not report their method of assigning participants to treatment and control groups.[29,30,32,33] Only five studies collected the participants' demographic information to compare homogeneity between treatment and control groups.[20,22,24,25,31] The eight other studies did not report the participants' demographic information and included this in their final analysis.

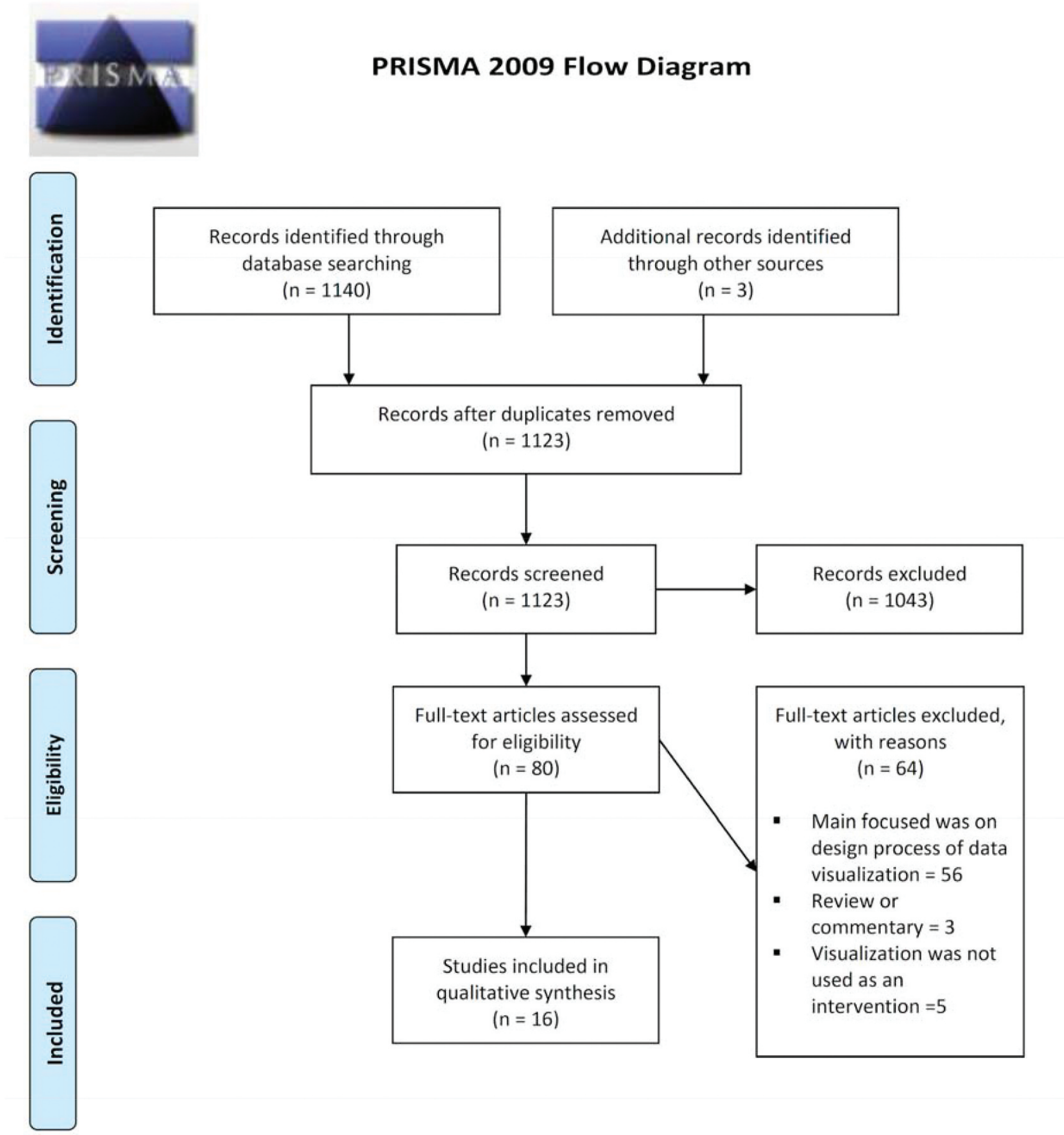

Figure 1. Flow diagram of included studies in a systematic review of data visualization interventions.

Participants

Sample sizes varied, ranging from 19 to 720 participants (mean = 140). The most common participants were volunteers from universities or institutions (e.g., a hospital) where the experiments were taking place.[20,28–30,32,34–36] Two studies recruited participants through Amazon Mechanical Turk, a crowdsourcing tool that enables individuals to perform tasks as paid work.[27,33] In three studies, the volunteer participants were the decision-makers whose work was closely related to the tested visualization tools (e.g., tracking ill patients' location in a hospital unit).[24,25,31] In two other studies, study participants were recruited from among high school students, either in the same grade or the same classrooms who were taking the same class or from university students selected to take a class in the same degree program.[21,23] This sampling method appeared as an effort to reduce heterogeneity between experimental and control groups.

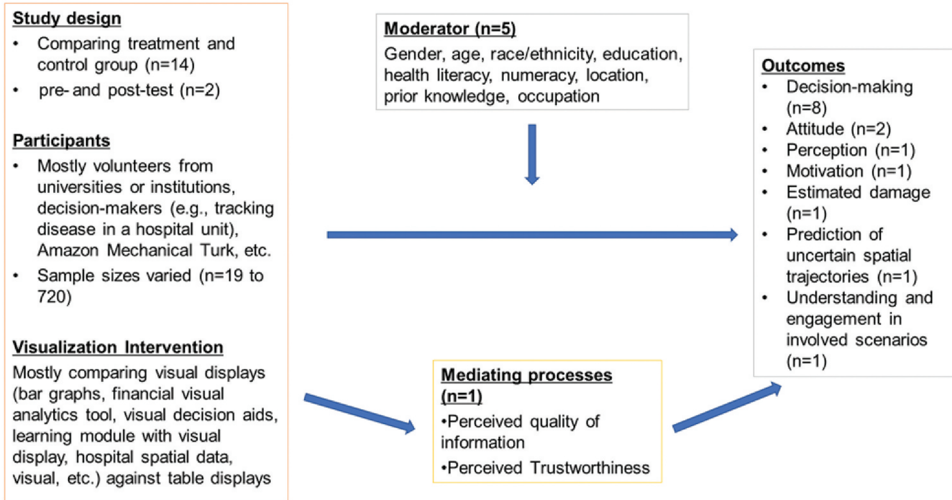

Figure 2. Summary of experimental and quasi-experimental studies, categorized by study design, intervention, participants, outcome measures, including mediating and moderating factors.

Most studies did not report how they calculated the appropriate sample size. Only two studies clearly stated the result of power analyses to calculate an appropriate sample size based on the study design.[22,31] Four studies compensated participants for completing tasks or study activities, ranging from 50 cents to $26, depending on the choice of their activities or how close they were to the optimal task result.[27,29,30,32,33]

Interventions

Interventions using data visualization varied widely across studies. Three studies compared the effects of the visually displayed information on decision-making or attitude change with the effects of textual information.[20,22,27] Four studies compared the effects of visual displays against tabular information for decision-making.[24,29–31] Two studies used an interactive visual tool that automatically sorts values on a spreadsheet as an intervention and compared how the tool influenced decision quality compared to a typical tool.[32,33] Of the two, the study by Kim added real-life scenarios (e.g., deciding on finding an apartment while considering its price, size, maintenance) in participants' decision-making processes. They asked them to consider them in that scenario. One study compared the effects of a decision-making learning module with a visualization tool, collaborative learning, and metacognitive guidance on decision-making by comparing pre-and posttest scores.[21] Two studies compared the effects of the visually displayed information on the prediction of a hurricane track or hypothetical trajectories.[35,36] One study compared an interactive web tool regarding an environmental scenario with a website with static scenarios and storylines.[25] Four studies held pre-treatments, including quiz sections or training activities for prerequisite skills before the experiments.[21,27,30,32] In Samek et al.'s study,[32] the quiz was put in place to determine participants' basic understanding of using the visualization tool. Control groups also varied across studies and included textual information only, baseline treatment with tabular information, or 2-dimensional images on PowerPoint slides.

Outcome measures

Outcome measures and mediating and moderating variables from the review are summarized in Table 2. Seven studies reported decision-making as a primary outcome.[21,22,24,29,30,32,33] One study reported the intention of use of visualized data for decision-making.[31] Two studies reported attitude changes as a primary outcome.[20,27] Another study reported on understanding and engagement in

Table 1. Study Designs and Intervention Characteristics in a Systematic Review of Data Visualization Intervention.

Study	Designs	Intervention; dosage	Control Group
Wang & Doong	experiment (n = 104);Randomized asTx group (n = 51) Control group (n = 53)	Participants watched an online advertising video and answered related questions. 1 session: 10 min (video: 3 min)compensation: not reported	Online information with text only
Rudolph et al.	Quasi-experiment (n = 26);Tx group (n = 13)Control (n = 13) (no randomization)	Participants used a visual analytics tool (FinVis), allowing the non-expert user to interpret features of financial data and make personal finance decisions. The tool helps the casual decision-maker quickly choose between various financial portfolio options and view possible outcomes. Seven sessions; duration of sessions not reported. Compensation: $9-18	Baseline treatment using a tabular display with the same information
Larson & Edsall	Experiment (n = 76);Randomized asTx group 1 (n = 35) Tx group 2 (n = 41)	Participants watched a 3D demonstration in Arizona State University's (ASU) Decision Theater (D.T.), featuring animated graphics surrounding viewers on large screen displays. The visual images in the D.T. primarily portrayed physical depth to the underground aquifer. Images on a peripheral screen – the 'digital dashboard'- consisted of a series of small supplemental graphs and legends to guide the presenter.The same person narrated, live, both presentations from a standard script. 10 min presentations addressed groundwater withdrawals while illustrating water levels and flow to pinpoint the cone of depression problems. compensation: not reported	2D PowerPoint presentation in a standard classroom, projected on a standard classroom screen, with smaller images in a narrower field of vision.
Westermann et al.	Experiment (n = 91);Randomized asTx group (n = 51) Control group (n = 45)	Visualization aids were used for creating a shared, understandable and meaningful narrative of diagnostic formulation and treatment options for parents of a child with a mental health problem (Counseling in Dialogue [CD]). Both CD and CU; 1 session, 60 mincompensation; not reported	Care as usual (C.U.)–parents are counseled in an equally long counseling session as in CD.
Samek et al.	Experiment (n = 120)Tx group 1Tx group 2Control group (randomization not reported)	Subjects were asked to make a decision, incentivized to select an object, and continue improving on their selection until they were satisfied with their selection, using three different treatments; Baseline, Typical Sort, and Automated Sort.One session; 140 min (20 rounds, 180 seconds for each round including instruction time) compensation; based on the selection of the final object as instructed average earning $26	Baseline; limited interactive capability to select and mouse-over objects

(*Continued*)

Table 1. (Continued).

Study	Designs	Intervention; dosage	Control Group
Pandey et al.	Experiment (n = 720);Randomized asTopic 1;Tx group (n = 54)Control group (n = 45)Topic 2;Tx group (n = 50)Control group (n = 56)Topic 3;Tx group (n = 55)Control group (n = 63)	9 stages that subjects participated in–demographic information, pre-treatment questions, persuasive message (context + evidence + presentation with graphical evidence (charts), post treatment questions, need for cognition scale)One treatment; 5–10 mincompensation: $ 0.50	Persuasive message by textual evidence (tables)
Savikhin et al.	Experiment (n = 56);Tx group 1 (n = 23)Tx group 2 (n = 14)Control group (n = 19)(randomization not reported)	Subjects took a pre-experiment quiz and were then exposed to interactive visual analytics and simple visual representations using 2D graphics to show the profit the subject would have received at every possible company value at their bid amount. After the outcome is determined, subjects may explore the dataset on their own time before making a final decision. compensation: $5-25	Table display
Anell, A., Hagberg, O., Liedberg, F., & Ryden, S.	Web-based survey Randomized as Tx group 1 (n = 113) Control group (n = 108)	Decision-makers compared hospital performance using either league tables or funnel plots for four measures of cancer care. For each measure, they were asked to suggest actions toward 12–16 hospitals and provide feedback on whether the information had been useful. Compensation: not reported	Data presentations in the form of League tables
Hsu, Y.-S., & Lin, S.-S.	Quasi-experiment (n = 74);pre-and posttests andRandomized as Tx group 1 (n = 42)Control group (n = 32)	Subjects participated in a decision-making learning module with three activities; 1) understanding the concept of evidence (reliability and validity of evidence), 2) Decision-making strategies, and 3) decision-making activities with/without embedded metacognitive guidanceOne learning module; 7 hoursCompensation; not reported	No metacognitive guidance
Velazquez-Iturbide, J. A., Hernan-Losada, I., & Paredes-Velasco, M.	experiment (n = 31);pre-and posttests andRandomized as Tx group 1 (n = 16)Control group (n = 15)	Participants used a program visualization system that displays recursive processes coded in Java to solve the class assignment. One session; 2 hours	Any Java programming tools the students had mastered (typically, BlueJ or Eclipse)
Kettelhut, V. V., Vanschooneveld, T. C., McClay, J. C., Mercer, D. F., Fruhling, A., & Meza, J. L.	Quasi-experimental survey (n = 19) with pre-and post-design	Participants viewed visualized hospital spatial data linked to individual patients' clinical-laboratory data for tracking ill patients' locations, antibiotic administrations, contact data, and infection prevention dataOne session; 30 minutes	Current EHR-based data
Kim S-H	Experiment (n = 127)Tx group (n = 67)Control group (n = 60) (randomization not reported)	The participants had three trials with different data sets using SimulSort for the interactive visualization interface. For each trial, a scenario was given. compensation; average earning $2.90	Typical Sorting for the nonvisual traditional interface

(*Continued*)

Table 1. (Continued).

Study	Designs	Intervention; dosage	Control Group
Ballard, A	Experiment (n = 319)Randomized as Tx group and control group (n for each group not reported)	Participants viewed a graph of an example indicator about performance related to restaurant inspections. Three sets of vignettes in 1 session	Table (numerical display) with the same data used for the graph (visual display)
Padilla, L. M. K., Creem-Regehr, S. H., & Thompson, W.	Experiment 1 (n = 200)Group 1 (9 tracks, n = 52)Group 2 (17 tracks, n = 50)Group 3 (33 tracks, n = 50)Group 4 (65 tracks, n = 48)	Participants were given visualizations of Hurricane Track Displays that mimicked distributions in such a way that one of the hurricane tracks passed through the center of a black dot (used to depict the "oil rig" location)	Group 1 (9 tracks) as a baseline
Pugh, A. J., Wickens, C. D., Herdener, N., Clegg, B. A., & Smith, C. A. P.	Experiment (n = 88)Randomized asTx group Control group(n for each group not reported)	Participants were tasked with predicting the location of various trajectories by calculating the absolute distance between the participant's predicted location and the model mean for each trial.Six experimental blocks (32 trials each block) and one practice block (16 trials)	The Control group viewed only one sample at a time and never viewed a representation of the data collectively
Xexakis, G., & Trutnevyte, E.	Experiment (n = 313)Randomized asTx group (n = 156) Control group (n = 157)	Participants use an interactive web tool with access to a large scenario database; illustrating 13 electricity supply alternatives in 2035, their constraints, and their environmental, health, and economic impacts	A website with four static scenarios and storylines

Table 2. Outcome measures, moderating, and mediating variables of studies to test visualizations.

Citation	Study Outcomes	Mediating Variables	Moderating Variables
Wang & Doong	The attitude of health care service (4 items with 4-point scale)	Perceived quality, Perceived Trustworthiness	Gender, age
Rudolph et al.	Decision-making (achieved a financial outcome) exploration and learning behavior (time spent) confidence and perception of usefulness (8-point scale)	Not reported	Not reported
Larson & Edsall	Perceptions about the magnitude of water resource problems and the causes of and solutions to water shortage (14 items with 5-point ordinal scales)	Prior knowledge (3 items with 5-point scale), Ecological worldviews; human control over nature, the fragility of the earth (6 items with 5-point scale),Political beliefs; free-market economy and private property rights, high values indicating conservative beliefs (2 items with 5-point scale)	Gender, age, race/ethnicity, location
Westermann et al.	Primary outcome: Decisional conflicts; uncertainty, factors contributing to uncertainty, and perceived effective decision-making (Decisional conflict scale, 16 items with 5-point Likert scale)Secondary outcome: Decision made upon accurate information, satisfaction with shared decision-making, consensus diagnostic formulation, consensus recommended treatment, and acceptance of recommended treatment (single item, 4-point scale)	Not reported	Gender, age, types of mental disease diagnosis, disease severity, education level of parents
Samek et al.	Decision-making process (achieved outcome value)	Not reported	Not reported
Pandey et al.	Persuasion likelihood (ratio of the number of persuaded participants and the total number of participants)Attitude change (single-item scales, changes in user's self-reported attitude)	Initial attitude before treatment (neutral or negatively polarized),degree of elaboration; involvement and need for cognition	Not reported
Savikhin et al.	Decision making achieved financial outcome value)	Not reported	Not reported
Anell, A., Hagberg, O., Liedberg, F., & Ryden, S.	Decision-making; Number of actions suggested by participants, the proportion of appropriate actions	Not reported	Age, gender, professional background, organizational position
Hsu, Y.-S., & Lin, S.-S.	Decision-making test and worksheets using a scoring rubric (four decision-making subskills scored from 0 to 2 points each)	Not reported	Not reported
Velazquez-Iturbide, J. A., Hernan-Losada, I., & Paredes-Velasco, M.	Students' motivation; the EMSI questionnaire was used to measure students' four dimensions of motivation	Not reported	Knowledge level: checked by grades obtained in previous assignments, the participants were fourth-year computer science major students who elected to take a course on advanced algorithms.

(*Continued*)

Table 2. (Continued).

Citation	Study Outcomes	Mediating Variables	Moderating Variables
Kettelhut, V. V., Vanschooneveld, T. C., McClay, J. C., Mercer, D. F., Fruhling, A., & Meza, J. L.	Perception and comprehension of the infection transmission risk factors using the VIZ vs. the current EHR-based data.	Not reported	Unit-based staff vs. non-unit-based staff (clinical consultants and infection preventionists)The novice (1–5 years in health care) vs. the experienced (>5 years in health care).
Kim S-H	Decision quality (whether dominated or nondominated option) and confidence level of the decision (a 7-point Likert scale), time spent making a decision	Not reported	Not reported
Ballard, A	likelihood of use of data for decision-making that displayed visually compared to the one displayed numerically (0–10 scale)	Not reported	Gender, race, age, experience, education's level
Padilla, L. M. K., Creem-Regehr, S. H., & Thompson, W.	Estimating damage that the platform will incur based on the depicted forecast of the hurricane path (7-point Likert scale, 1 is no damage and seven is severe). After each task, confidence in their predictions (7-point Likert scale, 1 (not at all confident) to 7 (very confident) for every trial)	Not reported	Not reported
Pugh, A. J., Wickens, C. D., Herdener, N., Clegg, B. A., & Smith, C. A. P.	The absolute distance (error) between the participant's predicted location and the model means for each trial, the probability that the target was located within the presented circle probes	Not reported	Not reported
Xexakis, G., & Trutnevyte, E.	Tested understanding (score range: 0–7), Self-reported understanding (score range: 6–42), Self-reported trust (score range: 7–49), Self-reported engagement (score range: 7–49), Tested engagement(time spent and drop-out rate in the website stage)	Not reported	Gender, education, Age, Prior interest and knowledge on topics, frequency of media use to inform on electricity supply topics, website navigation skills, subjective numeracy

involved scenarios.[25] The primary outcomes of the final four studies involved changes in perception on water resource problems,[28] motivation in computer programming tasks,[23] estimated damage occurred by a hurricane,[35] and the prediction of uncertain spatial trajectories.[36]

Decision-making

Among the studies that measured decision-making as a primary outcome, three studies asked participants to invest, allocate, or sort money using visualization aids to maximize returns.[29,30,32] The results of these studies were compared with control groups. Two other studies used either adapted scales such as the Decisional Conflict Scale (DCS) to measure the decision-making process,[22] or developed scoring rubrics such as decision-making tests and worksheets that score decision-making subskills from 0 to 2 points.[21] One study asked decision-makers working at administrative and clinical levels to compare hospital performance and suggest actions for hospital personnel by looking at either league tables (a widely used format of presentation) or funnel plots.[24] One study asked about the intended use of visualized data for decision-making related to restaurant inspection among local and county public health managers.[31] All findings in this review showed statistically significant effects on decision-making in the treatment groups compared to the control groups,[22,33] baseline information,[32] tabular information only,[27,29] and pre-and posttests.[21–23]

Attitude

Two studies reported a measure of attitude change using multi-item or single-item questionnaires as their primary outcome.[20,27] One study identified a significant difference in attitude toward healthcare services between a group that watched online healthcare advertising using a celebrity performing as a doctor showing both text and picture displays and a group that watched advertising using a celebrity performing as a patient showing text display only, mediated with perceived quality and trustworthiness of the advertisement.[20] The other study found mixed results for attitude change between a treatment and control group.[27] When the participants' initial attitudes were not polarized on socially controversial topics (e.g., the relationship between corporate income tax and unemployment rates in the U.S.), visual displays with charts had a stronger effect on the likelihood of attitude change and persuasion than displays that only used tabular information. When the participants' initial attitudes were polarized, the reverse trend was evident that participants shown the tabular display had a higher percentage of attitude change than participants shown the chart display.

Perception

One study reported perception change regarding water resource problems and solutions for a water shortage using a multi-item questionnaire with a 5-point Likert scale.[28] This study found both unchanged and changed perceptions depending on topics related to water management in response to visual information. The study also suggested that the changes in understanding or perception could be different depending on the types of information. For example, they found more highly significant changes in the perceived severity of water problems in the 2-D group (shown 2-D PowerPoint presentation) compared to the 3-D group (shown 3-D visualization).

Mediating and moderating factors

Five studies collected demographic information about the study participants.[20,22,24,25,31] The demographic information included gender, age, racial/ethnic background, education level, location, disease type and severity, prior interest and knowledge, health literacy, numeracy, professional background, and organization position. However, most studies did not include the demographic information in their analysis as a moderating factor; instead, they used moderating factors to check homogeneity between experiment and control groups. Only one study used gender as a moderating variable that strengthened the relationship of perceived quality and trustworthiness about visual advertisement content on attitude toward healthcare services.[20] Two studies found that initial attitude or prior knowledge of given topics significantly affected the strength of the relationship between the visual displays and the change in attitude or perception.[27,28] For example, in Larson and Edsall's study, prior knowledge about the water shortage problem, basic beliefs shared by social groups, and political beliefs were considered as exploratory factors, and it was found that greater knowledge of water issues was significantly associated with increased perceived water risk.[28] Only one study clearly articulated mediating factors. The study by Wang and Doong hypothesized that a treatment group that watches online healthcare advertising with both textual display and pictures would experience higher perceived quality and trustworthiness than a group only shown textual information. Their findings supported this hypothesis, indicating the mediating effects of perceived quality and trustworthiness on attitude toward healthcare services.[20]

Discussion

In this review, we assessed theories, methodological details, and results of 18 studies that aimed to examine how data visualization impacts attitude, motivation, perception, or decision-making. Although there were mixed results of the influence of data visualization on changes in cognition and behavior, systematic synthesis of the included studies showed that data visualization interventions have a positive impact on cognitive and behavior change such as decision-making (including the intention of using data for decision-making), attitude, motivation, understanding, engagement, or perception when compared to controls.[22,23,25,27,29–32,37] While our ultimate interest was concerning evidence for public health practice, the findings discussed here have relevance for a wide variety of fields and general evidence regarding the impact of data visualization.

Visualization appears to bring advantages by increasing the amount of information delivered and decreasing the cognitive and intellectual burden to interpret information for decision-making.[30] However, negative effects were also found. These negative relationships appear to be affected by moderating and mediating factors such as perceived trustworthiness and quality, domain-specific knowledge, basic beliefs shared by social groups, and political beliefs.[20,27,28] One study, for example, reported a negative effect of an interactive web tool as an intervention and speculated that interactive web tools might require a higher cognitive burden for participants, hampering performance in answering the requested task.[25] Issues associated with trust in displayed data were discussed in a previous study. Knowing the data source influenced how participants interpreted visual information due to limited trust regarding the visualization.[38]

Overall, evidence from this systematic literature review regarding the impact of data visualization was limited theoretically and methodologically. Most studies did not present theories used for their investigations, which would explain why and how data visualization tools work to impact cognitive processes and, as a result, influence decision-making behavior.

Due to the diversity of the academic disciplines and topics of included studies and the heterogeneous nature of data visualization tools, it was difficult to find consistency in methods and outcome measures from the findings. Unlike health science behavioral research, studies that test the impact of data visualization seem not to have either uniform protocols or guidelines for conducting behavioral studies, probably because the field is relatively new and has not yet begun to establish standardized protocols. Also, the heterogeneity of the studies may be because the visualization researchers have only recently realized the need for applying scientific methodologies, unlike researchers from cognitive psychology or social sciences who have been trained in the scientific method.[39,40] Most studies also did not appear to report the essential elements of an experimental study, which could jeopardize the perceived validity and reliability of the studies. For example, most studies did not report how they determined sample size or randomized participants into experimental and control groups. This issue was raised by Few in his critique as well.[40] Few criticized a study that tested the memorability of presented visualization by pointing out missing evidence regarding the determination of reliable sample size and, as a result, described a critical statistical flaw. Only a few studies considered potential confounding factors, including mediating and moderating factors, for their analyses. Incorporating individual differences such as intelligence, cognitive ability, and technical skill is important in designing studies because considering these individual characteristics in the early stages of studies would enable research to be more generalizable.[41] This systematic review suggests that gender, age, racial/ethnic background, educational level, professional background, health literacy, numeracy, place of living; as well as initial personal attitude, value judgments, and knowledge regarding data visualization seem to be possible mediating or moderating factors that should be considered in the assessment of the impact of data visualization.[20–22,24,27,28]

Differences in academic disciplines, target populations, aims, and methods across the study settings made this review challenging. Only one article was found to review the impact of data visualization on the audience of interest – public health leaders who make organizational decisions for programs and services in their agencies, for and with their communities. Nevertheless, the evidence from this review demonstrates the multiple benefits of data visualization and shows promise for its application to public health.

Building on the current literature, future studies should be designed with consideration of relevant theories and methodologies. Theories could be used to explain how visual displays help increase understanding of complex real-world problems using visual data and information from multiple sources resulting in effective and efficient decision-making. For example, in a study that describes a user's needs to develop a geo-visualization (GeoVis) framework, researchers emphasized incorporating cognitive fit theory into their visual geospatial displays combined with principles of public health and a human-centered approach.[42] The use of such theory explained how different graphical displays are related to user task performance and guided the researchers in designing a system that utilized the correct information based on the users' needs and the tasks they performed.[42]

Future studies should consider demographic characteristics specific to public health leaders. Koenig et al.[43] reported that public health practitioners who were novice users of geographic information struggled with box plots and required assistance to complete tasks. This may have been due to the lack of numerical training in public health curricula content.[43] Besides the range of demographic characteristics of individual public health leaders, researchers should understand in what contexts they use and understand data and how that informs the designing of visualization tools. Complete descriptions of tasks regarding how public health leaders make decisions for programs and services using data and data visualizations should precede the development of visualization tools. As discussed in our reviewed articles,[28,33] decisions were differently made when aided with data visualization. Due to the nature of public health systems, public health professionals' decisions are influenced by many other environmental and organizational factors such as policies, mandates, funding, organizational culture, and leadership.[6,44–48] Researchers could further investigate specific tasks that public health professionals perform and in what context their decision-making lies for assuring the best public health practice when they design and test data visualizations for public health professionals.

Strengths and limitations

This study is the first identified systematic literature review of the impact of data visualization on human behavior. Database searches were not limited to the discipline of public health but expanded to related disciplines such as computer science, information science, and public policy and included comprehensive searches of six relevant databases. Restrictions in the search strategies may have led to the exclusion of relevant studies.

Even though our search focused on experimental studies using data visualization to assess causal impact, qualitative studies or usability studies may also be important to understand the use of data visualization for public health leaders and others. Finally, a meta-analysis of the primary studies could not be performed due to heterogeneity and a small number of studies.

Conclusion

This review assessed theories, methodological details, and results of 16 quasi- and experimental studies that aimed to examine the impact of interventions incorporating data visualization. Even with increasing interest in visualization evaluation studies in recent years,[49] there is still relatively little research on how visualization impacts decision-making, including cognitive and behavioral processes, especially in the discipline of public health.

The evidence from this review generally suggests positive effects of data visualization depending on the control of confounding factors on behavioral changes such as attitude, perception, and decision-making. However, understanding regarding data visualization interventions specific to public health leaders' decision-making still lacks, in addition to which there is little guidance to understand participants' characteristics and tasks. A visualization study that involves public health leaders from the beginning of the design process could aid in the development of effective visualization tools and interventions that accurately depict real-world problems and support the needs of public health leaders. Future research to examine the impact of data visualization should be conducted with the appropriate methodology, specifically targeting public health leaders to improve the quality of evidence.

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APPENDIX

Literature searches started by breaking down the research question and conducting scoping searches on the PubMed database to identify search terms according to PICOS (Population, Interventions, Comparisons, Outcomes, and Study Design).

- Research question: Data visualization as a behavioral intervention for policy-related behavior changes in state and local public health
- Population(s)/Patient(s): public health leaders
- Intervention(s)/Treatment(s): data visualization
- Comparator(s): table form of result
- Outcome(s): policy-related behavior change
- Study Design: Any type of study design
- Setting: community and public health system, or state and local health department

Next, using the developed search strategies, searches were conducted on PubMed, CINAHL Plus with full text, Web of Science, IEEE Xplore, PAIS international, and Academic Search Complete and retrieved primary studies. The publication period ranged from 2008 to 2020. Foreign language papers were excluded. 358 articles from PubMed, 78 articles from CINAHL Plus with full text, 303 articles from Web of Science, 244 articles from IEEE Xplore, 76 articles from PAIS international, and 81 articles from Academic Search Complete were found. Three additional articles were identified by reviewing reference lists of highly relevant articles. Overall, a total of 1123 articles were found (20 including duplicates). Concept tables were constructed as I proceeded with the search on the database.

Concept Table

PUBMED	Concept: VDQI/data visualization	Concept: policy-related behavior	Concept: public health
Mesh Terms	Visual display of quantitative information Data visualization Information visualization	Health planning Health Planning Organizations Decision-making Decision Making, Organizational Policy Making Decision Support Techniques Behavior Information Seeking Behavior/ subheading Attitude Attitude to Computers/ subheading Comprehension	Public health Community health Public Policy Health policy/ subheading Administrative personnel
CINAHL Plus	**Concept: VDQI/data visualization**	**Concept: policy-related behavior**	
CINAHL Headings	Visual display of quantitative information Data visualization Information visualization	Health and Welfare Planning Health Services Needs and Demand Marketing National Health Programs State Health Plans Strategic Planning Decision making, organizational Explode Decision Support Techniques Explode Problem Solving Health Service Administration Information management Explode Knowledge Management Nursing Administration Organizational Policies Planning Techniques Explode Resource Allocation Shared Governance/subheading Policy Making Public Policy Behavior Attitude understanding	
Web of Science	**Concept: VDQI/data visualization**	**Concept: policy-related behavior**	**Concept: public health**
	Visual display of quantitative information Data visualization Information visualization	Health planning Decision-making Policy Making Decision Support Behavior Attitude Comprehension	Public health Community health Public Policy Administrative
PAIS international	**Concept: VDQI/data visualization**	**Concept: policy-related behavior**	
	Visual display of quantitative information Data visualization Information visualization	Health planning Decision making Policy Making Decision Support Behavior Attitude Comprehension Understanding	
Academic Search Complete	**Concept: VDQI/data visualization**	**Concept: policy-related behavior**	

(*Continued*)

(Continued).

	Visual display of quantitative information Data visualization Information visualization	Health planning Decision making Policy Making Decision Support Behavior Attitude Comprehension Understanding
IEEE Xplore	**Concept: VDQI/data visualization**	**Concept: policy-related behavior**
	Visual display of quantitative information Data visualization Information visualization	Health planning Decision making Policy Making Decision Support Behavior Attitude Comprehension Understanding